CONNECT & SIMPLIFY

ALLSELL System Design & Product Architecture

Comprehensive multi-role system specification for Ethiopia's dual-vertical marketplace unifying **Houses (Real Estate)** and **Cars (Automotive)** with verified broker KYC, Telebirr lead monetization, and an institutional Government Auditor portal.



Client: Nishank / ALLSELL Team **Version:** 1.0 Final Specification

Platform: Flutter (iOS & Android) + Node.js

Market: Addis Ababa & Regional Ethiopia

1 Executive Summary & Brand Identity

ALLSELL is an end-to-end e-commerce and classifieds marketplace platform tailored specifically for the Ethiopian market. While existing platforms are either informal Telegram channels or single-vertical portals, ALLSELL combines the high discovery speed of *Gari / Afrotie* with the accessible e-commerce simplicity of *Tinamart*.

> **Core Value Proposition:** Eradicating counterfeit listings and broker fraud in Ethiopia by enforcing mandatory National ID & TIN verification, issuing cryptographic trust badges, and monetizing buyer inquiries through automated Telebirr and CBE Birr micro-transactions.

Official Brand Color Palette

Primary Amber
`#FF8D00`

Deep Obsidian
`#000000 / #0B0E14`

Warm Surface Tint
`#F7F5F0 (Wireframe tone)`

Verified Green
`#15803D (Trust Badge)`

2 The Four Core Business Pillars

The entire platform architecture rests upon four institutional pillars designed for scalability and compliance:



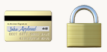
1. Unified Dual Marketplace

- Houses (Villas, Apartments, Condos for Rent & Sale).
- Cars (Brand New, Ethiopian Used, Foreign Imports).
- Sub-city location routing (Bole, CMC, Kirkos, Yeka).
- Instant category tab switcher with zero reload lag.



2. Verified Broker Ecosystem

- Mandatory KYC: National ID card & TIN Certificate.
- Business license validation where applicable.
- Moderator review before issuing the "✓ Verified" mark.
- Broker performance and lead tracking CRM dashboard.



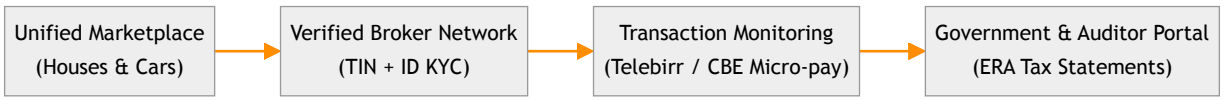
3. Transaction & Lead Monitoring

- Pay-per-contact reveal: Gated behind Telebirr / CBE Birr.
- Filters out casual window-shoppers & lead scrapers.
- End-to-end transaction logging (Initiated -> Paid -> Closed).
- Platform commission split automation.



4. Government & Auditor Portal

- Unique regulatory portal for Ethiopian Revenue Authority.
- Real-time gross sales and rental volume tracking (ETB).
- Automated broker income statements and tax reports.
- Immutable audit logs for dispute resolution.


Unified Marketplace
(Houses & Cars)
Verified Broker Network
(TIN + ID KYC)
Transaction Monitoring
(Telebirr / CBE Micro-pay)
Government & Auditor Portal
(ERA Tax Statements)

3 User Roles & Access Control Matrix (RBAC)

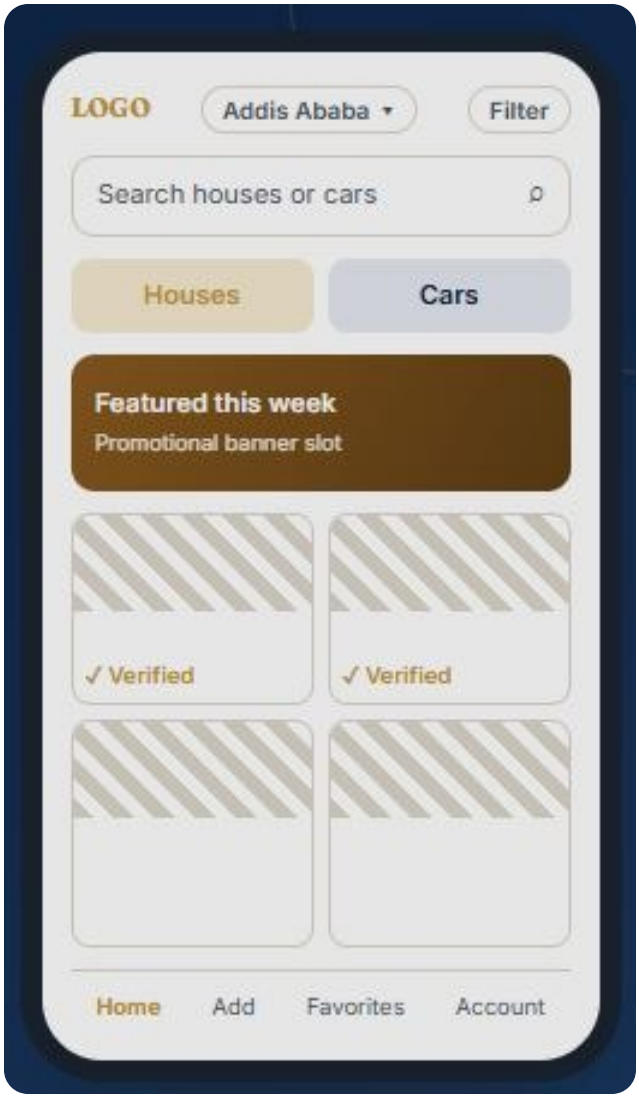
ALLSELL implements 6 distinct user roles to guarantee data privacy and regulatory separation:

ROLE	PERMISSIONS & SCOPE	MONETIZATION & AUTH	TARGET CLIENT SURFACE
CUSTOMER	Browse, search, filter listings, save favorites, make inquiries, report fake listings.	Phone + OTP auth. Pays nominal fee (20 ETB) to view broker contact details.	Mobile App (iOS / Android)
AGENT / BROKER	Create & manage house/car listings, view lead inquiries, manage customer CRM, view commission payouts.	Full KYC (TIN, National ID, Phone 1 & 2, Photo). Receives verified badge upon approval.	Mobile App (Agent Mode) & Broker Web
MODERATOR	Review pending listings, authenticate submitted ownership and TIN documents, approve/reject/suspend.	Internal staff auth with multi-factor authentication.	Staff Moderator Web Dashboard
ADMINISTRATOR	Manage platform categories, system settings, promotional banner slots, commission rates, and dispute resolution.	High-security privileged access.	Executive Admin Panel

ROLE	PERMISSIONS & SCOPE	MONETIZATION & AUTH	TARGET CLIENT SURFACE
SUPER ADMIN	Assign admin privileges, trigger database snapshots and system backups, configure payment API credentials.	Root system permissions.	Master DevOps / Root Console
AUDITOR / GOV	Read-only compliance dashboard: Agent registration statistics, gross sales (ETB), rental turnover, tax withholding statements.	Authorized government SSO / Token with audit trail.	Dedicated Government / Auditor Portal

4 Mobile App UI/UX & Wireframe Specification

The mobile experience directly mirrors the wireframe mockup, optimizing for single-handed usage on modern smartphones while keeping the core navigation instantly understandable.



OFFICIAL ALLSELL MOBILE WIREFRAME

1. Header Bar: Brand Logo & Location Selector

Prominent **ALLSELL Logo** on the left, an interactive **Addis Ababa** ▾ sub-city selector pill in the center (Bole, CMC, Kirkos, Yeka, etc.), and a dedicated **Filter** icon on the right.

2. Universal Smart Search

Contextual search field with placeholder *"Search houses or cars"*. Dynamically adapts suggestions based on the active tab (e.g., "Toyota Land Cruiser" vs "Bole 3 Bed Apartment").

3. Dual-Category Segmented Switcher

Two tactile pill buttons: **[Houses]** and **[Cars]**. Tapping triggers a smooth micro-animation and instantaneously switches the underlying feed and filter dimensions.

4. Promotional Banner Slot ("Featured this week")

A high-impact amber-brown gradient banner card. Used for monetized sponsored listings, verified dealership spotlights, or mortgage/loan partnerships.

5. Two-Column Responsive Listing Grid

Optimized 2-column card layout: photo thumbnail, price in

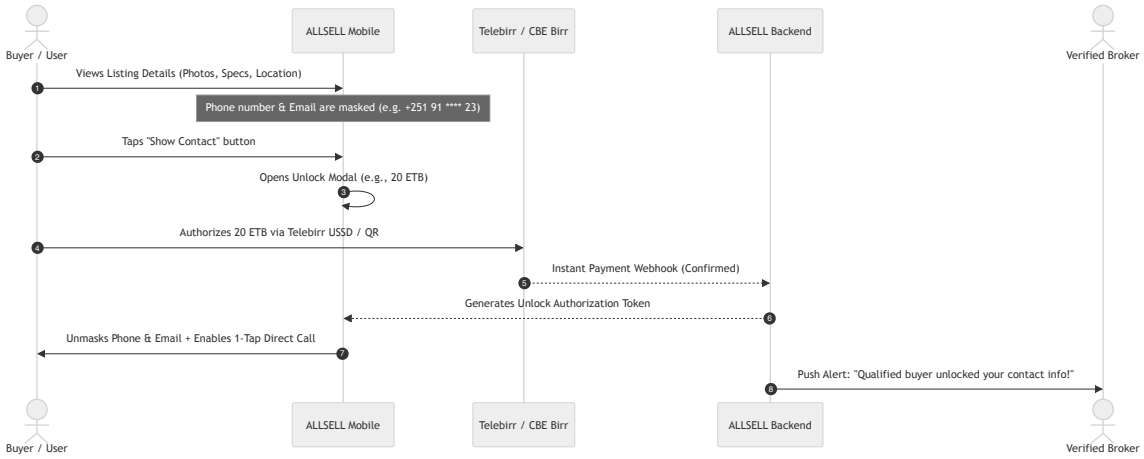
ETB, title, key specs chips, and the prominent ✓ **Verified** badge in gold/amber.

6. Bottom Navigation Bar (4 Touchpoints)

Clean 4-destination bar: **Home** (Discovery), **Add** (Sell/List Wizard), **Favorites** (Saved items), and **Account** (Profile & Agent Portal).

5 Lead Monetization & Telebirr Contact Paywall

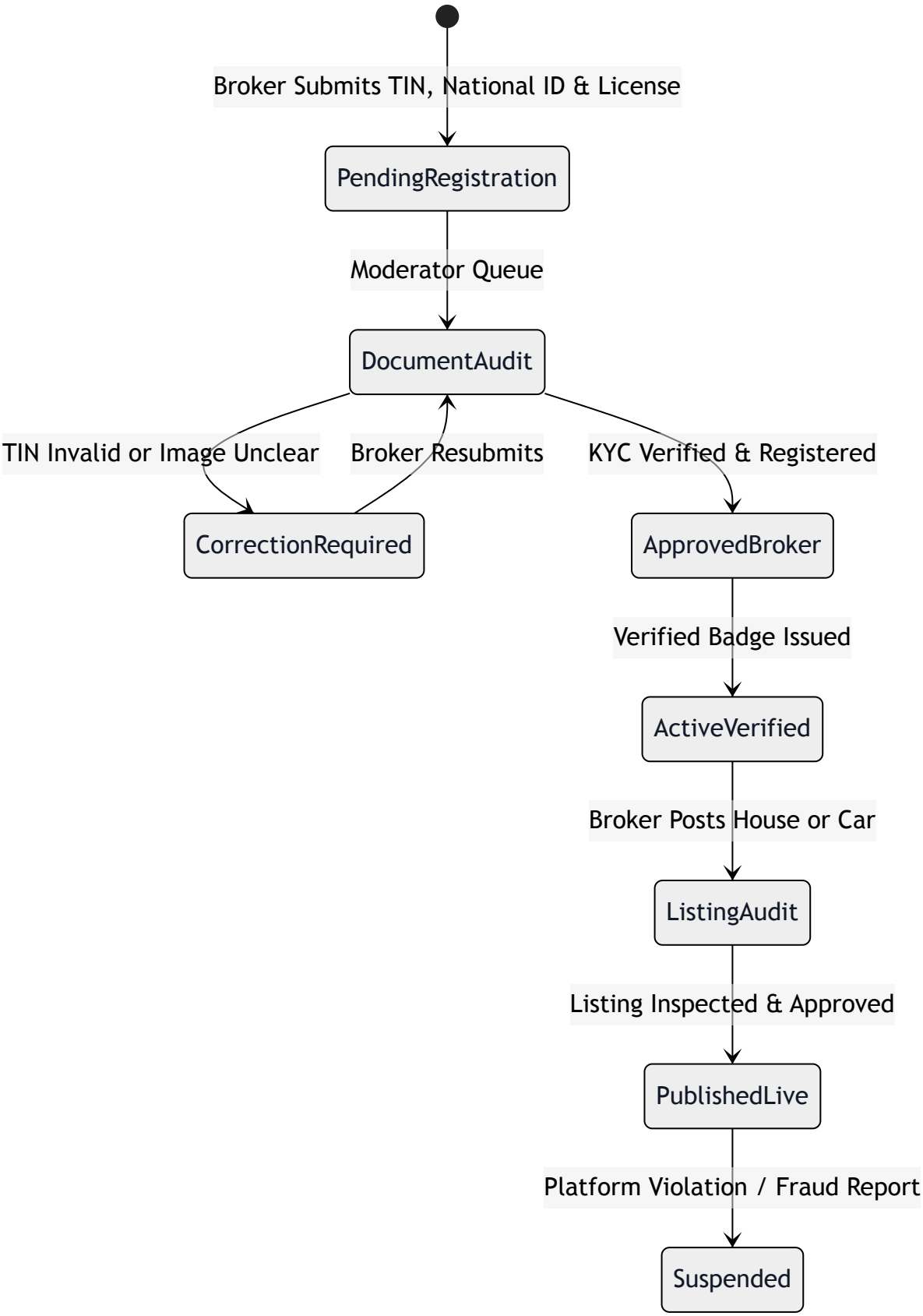
In traditional classifieds, phone numbers are scraped and brokers receive endless spam. ALLSELL introduces the **Pay-to-Reveal Contact Gate**:



Business Impact: Generates continuous platform revenue on high-frequency buyer intent, ensures only serious buyers reach brokers, and establishes a quantifiable lead conversion pipeline.

6 Agent KYC & Listing Verification Pipeline

To guarantee that only legitimate properties and vehicles appear on the platform, a four-stage audit pipeline is enforced:


Broker Submits TIN, National ID & License
PendingRegistration
Moderator Queue
DocumentAudit
TIN Invalid or Image Unclear
Broker Resubmits
KYC Verified & Registered
CorrectionRequired
ApprovedBroker
Verified Badge Issued
ActiveVerified
Broker Posts House or Car
ListingAudit
Listing Inspected & Approved
PublishedLive
Platform Violation / Fraud Report
Suspended

7 Government & Auditor Compliance Portal

A standout differentiator mandated in the system requirements is the **Dedicated Regulatory Portal** for authorized Ethiopian government bodies and external auditors.

DASHBOARD METRIC	SAMPLE VALUE	COMPLIANCE PURPOSE
Registered Brokers	1,250 Agents	Monitoring formalization of informal real estate & automotive brokers.
Active Verified Brokers	980 Agents	Brokers with active TIN certificates and verified transactions.
Total Completed Transactions	8,540 Deals	Platform-wide volume for the fiscal period.
Total Gross Sales Value	ETB 450,200,000	Aggregate property and vehicle transaction capital flow.
Total Gross Rental Value	ETB 82,400,000	Aggregate residential and commercial lease volume.
Broker Commissions Earned	ETB 13,506,000	Gross broker earnings utilized for calculating estimated tax withholding.

** Note: All government access is read-only, cryptographically logged, and compliant with Ethiopian data privacy regulations.*

8 Database Models & Entity Architecture

The backend separates real estate domain specifics from vehicle specifics while unifying them under shared transaction, media, and authorization tables:

```
// 1. PROPERTIES SCHEMA (Houses & Real Estate)
Table properties {
  id: uuid [pk]
  agent_id: uuid [ref: > users.id]
  listing_type: enum('FOR_SALE', 'FOR_RENT')
  property_type: enum('VILLA', 'APARTMENT', 'CONDOMINIUM', 'COMMERCIAL', 'LAND')
  title: varchar
  price: decimal(14,2)
  currency: varchar(3) // 'ETB'
  region: varchar // 'Addis Ababa'
  sub_city: varchar // 'Bole', 'Kirkos', 'Yeka', etc.
  wereda: varchar
  bedrooms: int
  bathrooms: int
  area_sqm: decimal(8,2)
  furnishing_status: enum('FURNISHED', 'UNFURNISHED', 'SEMI')
  utilities: jsonb // { water: true, backup_generator: true, electricity: true }
  is_verified: boolean
  status: enum('ACTIVE', 'PENDING', 'SOLD', 'RENTED')
}

// 2. VEHICLES SCHEMA (Cars & Automotive)
Table vehicles {
  id: uuid [pk]
  agent_id: uuid [ref: > users.id]
  make: varchar // 'Toyota'
  model: varchar // 'RAV4'
  year: int // 2023
  price: decimal(14,2)
  currency: varchar(3) // 'ETB'
  condition: enum('BRAND_NEW', 'ETHIOPIAN_USED', 'FOREIGN_USED')
  mileage_km: int
  transmission: enum('AUTOMATIC', 'MANUAL')
  fuel_type: enum('PETROL', 'DIESEL', 'ELECTRIC', 'HYBRID')
  body_type: enum('SUV', 'SEDAN', 'PICKUP', 'HATCHBACK')
  is_verified: boolean
```

```
  status: enum('ACTIVE', 'PENDING', 'SOLD')
}
```

9 Execution & Migration Roadmap

Evolution of the current workspace into the production ALLSELL platform:

Phase 1: Design & Client Architecture Sign-Off (Current Step)

System design specification, wireframe harmonization, database models, and client alignment.

Phase 2: Visual Identity & Color Theme Rebrand

Apply Amber (#FF8D00), Obsidian Black (#000000), warm off-white, and official AllSell branding across Flutter theme.

Phase 3: Wireframe Home Screen Implementation

Build header with Addis Ababa sub-city picker, search bar, [Houses] & [Cars] dual tabs, promotional banner slot, and 2-column verified card grid.

Phase 4: Real Estate & Automotive Detail Screens with Paywall

Dual detail templates with photo galleries, Ethiopian utility chips (water, generator), and Telebirr contact unlock modal.

Phase 5: Agent Dashboard & Listing Creation Wizard

Agent KPI metrics (Total listings, sales ETB, commissions) and KYC document upload interface.

Phase 6: Auditor Portal & Backend API Integration

Node.js / Express backend with PostgreSQL, simulated Telebirr webhook, and read-only auditor reporting views.

Project Approval & Sign-Off

By signing below, the client and development team approve this System Design Document as the architectural foundation for ALLSELL Phase 1 implementation.

FOR THE CLIENT (ALLSELL / NISHANK):

Authorized Signatory / Date

FOR THE ENGINEERING TEAM:

Lead Architect / Date